

Elasticity

**ECON 101: Introduction to
Microeconomics**

Prof. Alejandro Melo Ponce



Elasticities of Supply and Demand

Elasticity: $\frac{\Delta \% Y}{\Delta \% X}$
 (of Y with respect to X)

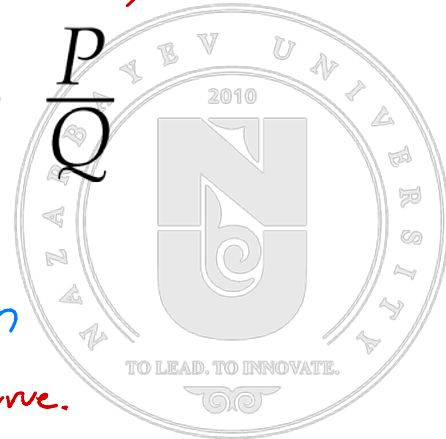
- **Elasticity** Percentage change in one variable resulting from a 1-percent increase in another.
- **Price Elasticity of demand** Percentage change in quantity demanded of a good resulting from a 1-percent increase in its price.

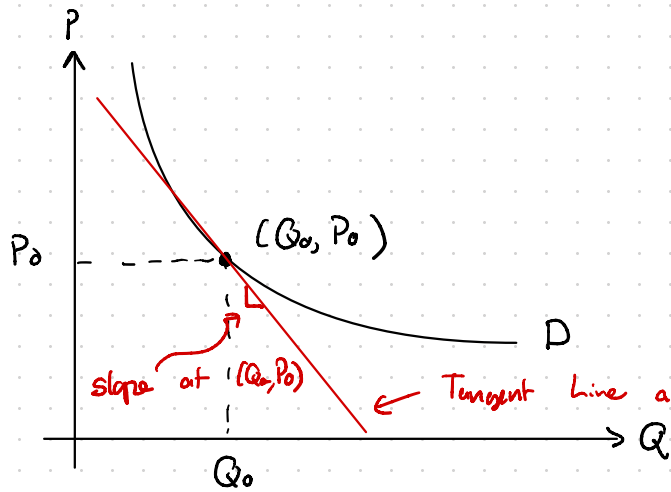
$$\epsilon_p^D = \frac{\% \Delta Q}{\% \Delta P} = \frac{\Delta Q / Q}{\Delta P / P} = \overset{(1)}{\frac{\Delta Q}{\Delta P}} \cdot \overset{(2)}{\frac{P}{Q}}$$

Point-price elasticity of demand:

(1) $\frac{\Delta Q}{\Delta P}$ needs to be computed from the demand schedule function

(2) $\frac{P}{Q}$ we just need a point (P, Q) in the demand curve.





For a non-linear demand curve you need either

- the exact demand function

OR

- another point to approximate the slope.



points to a limitation of the point-price elasticity of demand:

ONLY accurate when the $\% \Delta P$ is small!

Consider that $Q_D = a - bP$, $b > 0$, $a > 0$

suppose $P_1 \rightarrow P_2$:
 P_1 : $Q_1 = a - bP_1$
 P_2 : $Q_2 = a - bP_2$

$$\Rightarrow \Delta Q = Q_2 - Q_1 = a - bP_1 - (a - bP_2) \\ = -b(P_2 - P_1)$$

$$\Delta P = P_2 - P_1$$

$$\therefore \frac{\Delta Q}{\Delta P} = \frac{-b(P_2 - P_1)}{P_2 - P_1} = -b$$

Point-Price elasticity for a Linear Demand Curve

$$\epsilon_P^D = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q} = -b \cdot \frac{P}{Q} = \frac{-b \cdot P}{a - bP}$$

at point (Q, P)

Obs 1: Notice that ϵ_P^D is non-constant
for a linear demand curve

Obs 2: $\epsilon_P^D < 0$, (Law of Demand)

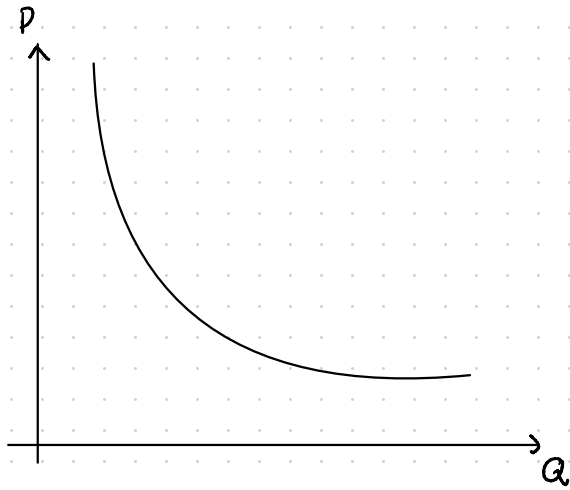
$\Rightarrow$ useful in terms of $|\epsilon_P^D|$

For any general demand curve:

$|E_p^D| > 1$: demand is elastic $\frac{1\% \Delta Q}{1\% \Delta P} > 1 \Rightarrow 1\% \Delta Q > 1\% \Delta P$
quantity changes more than
the price changes

$|E_p^D| < 1$: demand is inelastic : $\frac{1\% \Delta Q}{1\% \Delta P} < 1 \Rightarrow 1\% \Delta Q < 1\% \Delta P$
quantity changes less than
the price change

$|E_p^D| = 1$: demand unitary elastic : $\frac{1\% \Delta Q}{1\% \Delta P} = 1 \Rightarrow 1\% \Delta Q = 1\% \Delta P$
quantity changes equally to
the change in price.



$$Q_D = \frac{A^D}{P}$$

Demand eq

$$\Rightarrow P = \frac{A^D}{Q_D}$$

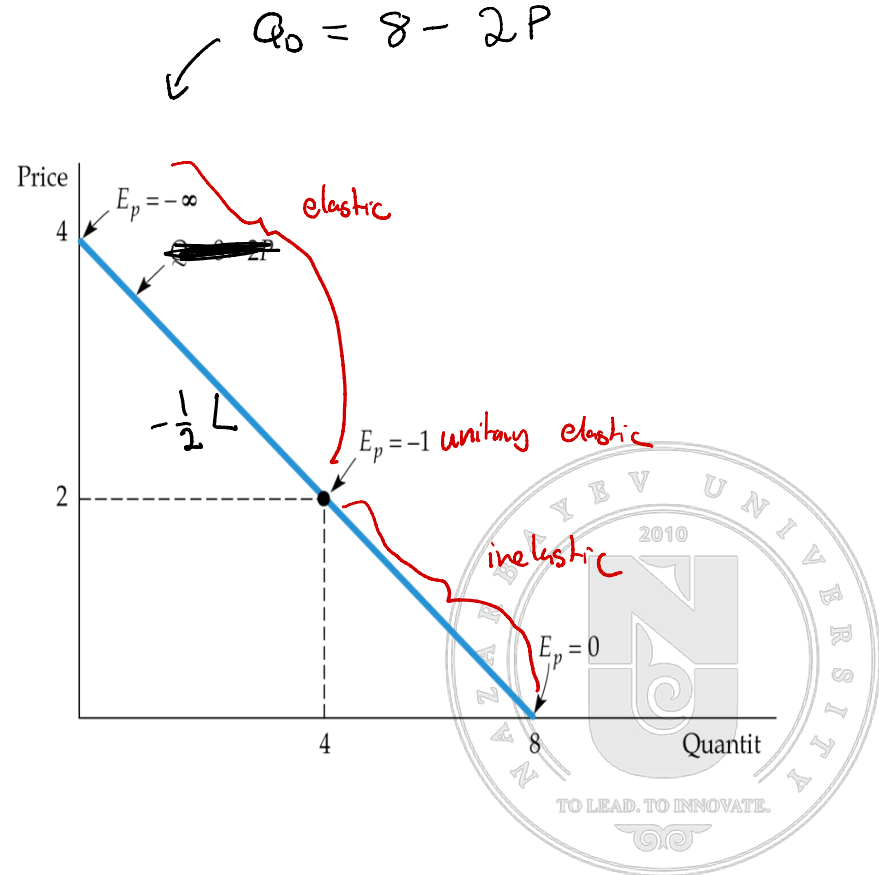
Inverse demand

$$|\epsilon_D^D| = 1 \quad \text{for any } (Q, P)$$

Elasticities of supply and Demand

Linear Demand Curve

- The price elasticity of demand depends not only on the slope of the demand curve but also on the price and quantity.
- The elasticity, therefore, varies along the curve as price and quantity change. Slope is constant for this linear demand curve.
- Near the top, because price is high and quantity is small, the elasticity is large in magnitude. The elasticity becomes smaller as we move down the curve.



To understand the regions of a linear demand curve:

- suppose you were a single seller, facing the demand curve

$$Q_D = 8 - 2P$$

↳ as the single seller you are able to set the price

What are your revenues: $R(Q) = \underline{P} \cdot Q$

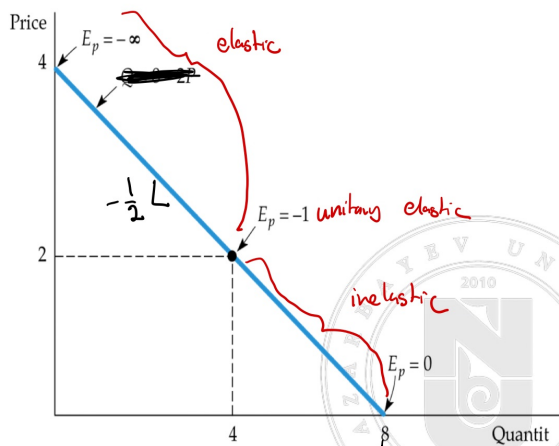
P is a "function of Q "

$$= P(Q) \cdot Q$$

Inverse demand

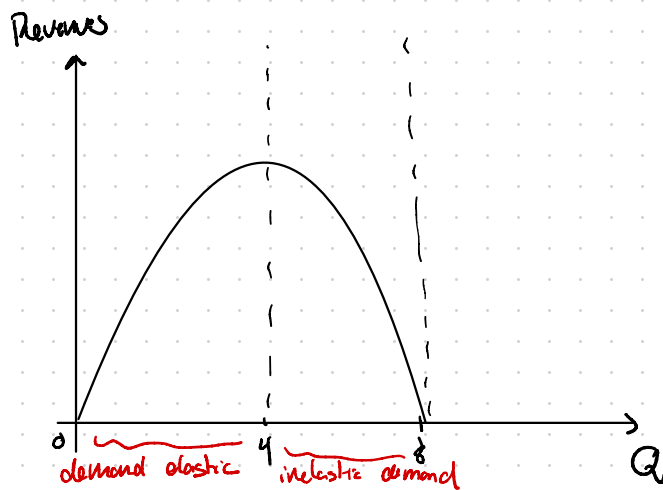
$$P(Q) = 4 - \frac{1}{2}Q$$

$$= (4 - \frac{1}{2}Q)Q = 4Q - \frac{1}{2}Q^2$$



$$P(Q) = 4Q - \frac{1}{2}Q^2$$

$$= (4 - \frac{1}{2}Q)Q$$



Elasticities of Supply and Demand

(a) INFINITELY ELASTIC DEMAND

For a horizontal demand curve, $\Delta Q/\Delta P$ is infinite. Because a tiny change in price leads to an enormous change in demand, the elasticity of demand is infinite.

infinitely elastic demand Principle that consumers will buy as much of a good as they can get at a single price, but for any higher price the quantity demanded drops to zero, while for any lower price the quantity demanded increases without limit.

For a perfectly elastic demand curve:

$$P = P^*$$

$$Q = \begin{cases} [0, \infty) & \text{if } P = P^* \\ 0 & \text{otherwise} \end{cases}$$



(a)



Elasticities of Supply and Demand

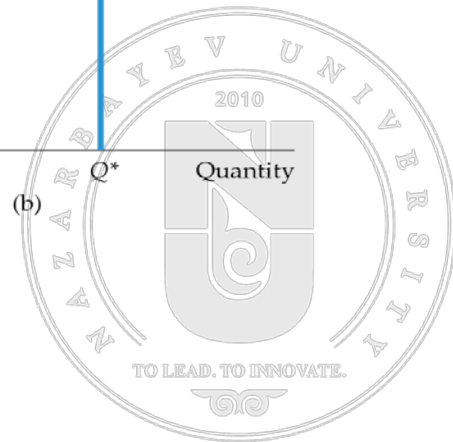
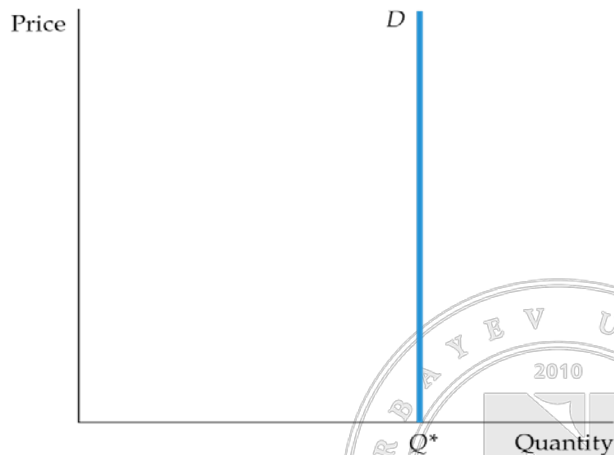
(a) PERFECTLY (COMPLETELY) INELASTIC DEMAND

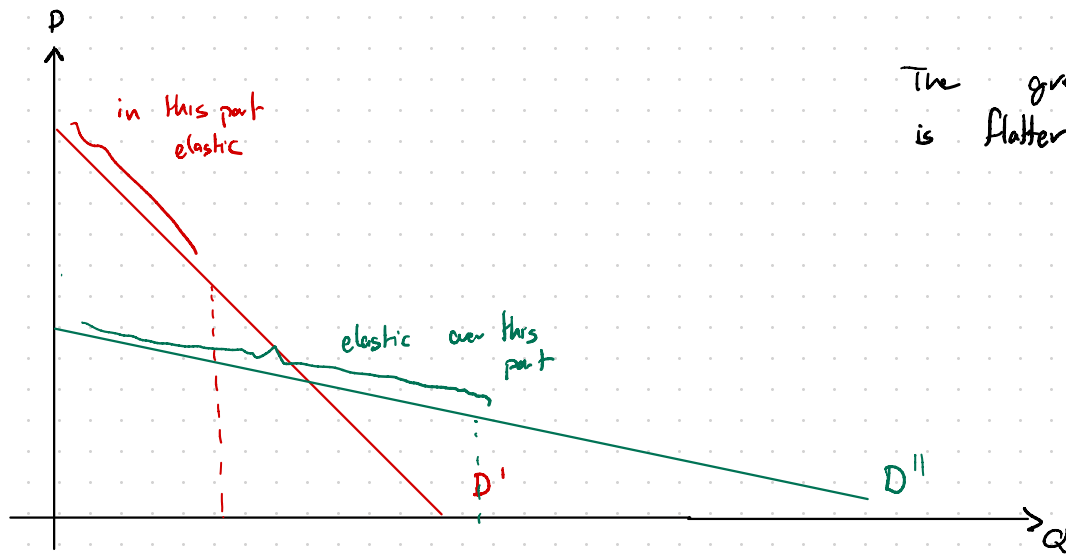
For a vertical demand curve, $\Delta Q / \Delta P$ is zero. Because the quantity demanded is the same no matter what the price, the elasticity of demand is zero.

Perfectly inelastic demand Principle that consumers will buy a fixed quantity of a good regardless of its price.

$$Q_D = Q^*$$
$$P = \begin{cases} [0, \infty) & \text{if } Q = Q^* \\ 0 & \text{otherwise} \end{cases}$$

inverse demand





The green one, since it is flatter is more elastic.

Elasticities of Supply And Demand

OTHER DEMAND ELASTICITIES

Income elasticity of demand Percentage change in the quantity demanded resulting from a 1-percent increase in income.

$$\epsilon_I^D = \frac{\% \Delta Q}{\% \Delta I} = \frac{\Delta Q / Q}{\Delta I / I} = \frac{\Delta Q}{\Delta I} \cdot \frac{I}{Q}$$

Cross-price elasticity of demand

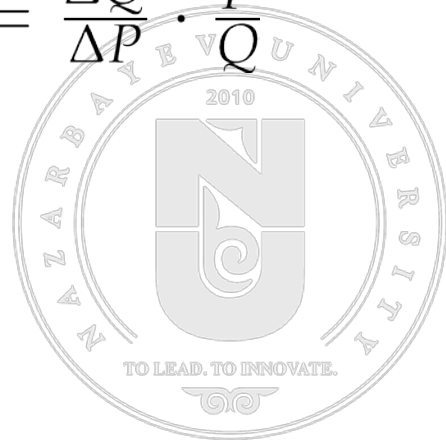
Percentage change in the quantity demanded of one good resulting from a 1-percent increase in the price of another.

$$\epsilon_{Q_b, P_m} = \frac{\% \Delta Q_b}{\% \Delta P_m} = \frac{\Delta Q_b / Q_b}{\Delta P_m / P_m} = \frac{\Delta Q_b}{\Delta P_m} \cdot \frac{P_m}{Q_b}$$

ELASTICITIES OF SUPPLY

Price elasticity of supply Percentage change in quantity supplied resulting from a 1-percent increase in price.

$$\epsilon_p^S = \frac{\% \Delta Q}{\% \Delta P} = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$



Example: $Q_0 = a - bP + fI + gP_s - hP_c$

where a, b, f, g, h are strictly positive constants.

$$\frac{\Delta Q}{\Delta I} = \frac{Q_2 - Q_1}{I_2 - I_1} = f ; \quad \frac{\Delta Q}{\Delta P_s} = g ; \quad \frac{\Delta Q}{\Delta P_c} = -h$$

$$\epsilon_I^D = \frac{\Delta Q}{\Delta I} \cdot \frac{I}{Q} = f \cdot \frac{I}{Q} = f \frac{I}{a - bP + fI + gP_s - hP_c}$$

$$\epsilon_{Q_0, P_s} = \frac{\Delta Q_0}{\Delta P_s} \cdot \frac{P_s}{Q_0} = g \frac{P_s}{Q_0}$$

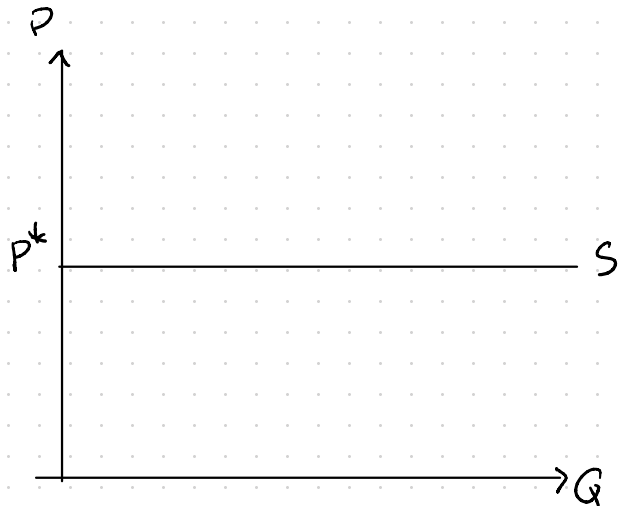
$$\epsilon_{Q_0, P_c} = \frac{\Delta Q_0}{\Delta P_c} \cdot \frac{P_c}{Q_0} = -h \frac{P_c}{Q_0}$$

$$Q_s = c + dP$$

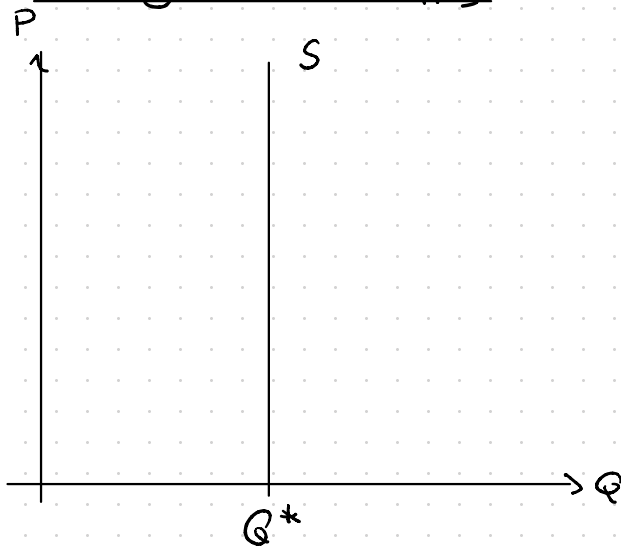
$$\frac{\Delta Q_s}{\Delta P} = d$$

$$\begin{aligned} \epsilon_P^S &= \frac{\Delta Q_s}{\Delta P} \cdot \frac{P}{Q_s} = d \frac{P}{Q_s} \\ &= d \frac{P}{c + dP} \end{aligned}$$

Perfectly Elastic Supply



Perfectly Inelastic Supply



Example:

$\swarrow$ millions of lbs / month

$$Q_D = 104 - 40p + 20p_t + 0.01Y$$

$\hookrightarrow$ Market for Avocados

p = price of avocados

p_t = price of tomatoes

Y = Income

$$Q_D' = 104 - 40p + 20p_t + 0.01Y'$$

$$\Delta Q_D = Q_D' - Q_D = 0.01(Y' - Y) = 0.01 \Delta Y$$

$$\hookrightarrow \frac{\Delta Q_D}{\Delta Y} = 0.01$$

How would Q_D change after a change in Y of 1% : $\Delta Y\% = 0.01$

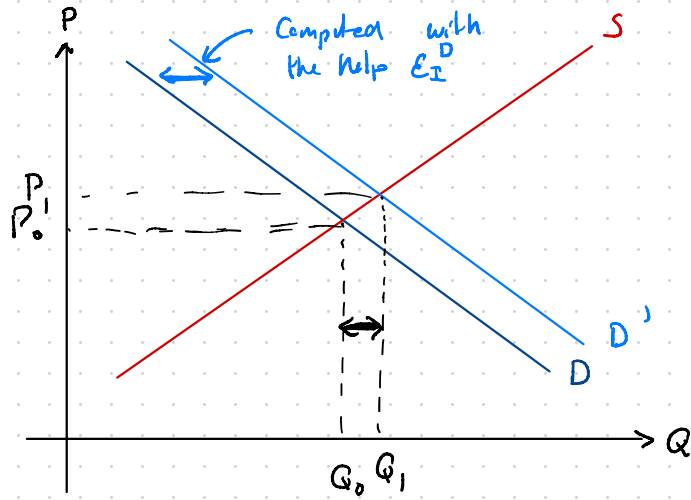
$Q = 50$, $Y = 4,000$, $p = 2$, p_t fixed.

$$\mathcal{E}_I^D (Q=50, Y=4,000) = \frac{\Delta\% Q_D}{\Delta\% Y} = \frac{\frac{\Delta Q_D}{Q}}{\frac{\Delta Y}{Y}} = \frac{\Delta Q_D}{\Delta Y} \cdot \frac{Y}{Q} = 0.01 \times \frac{4,000}{50} = 0.8$$

If $\Delta\% Y = 0.01$ or 1% $\Rightarrow \Delta Q_D\% = 0.8$ or 0.8%

$$\mathcal{E}_I^D = \frac{\frac{\Delta Q_D}{Q}}{\Delta Y\%} \Rightarrow \begin{array}{ccccc} \mathcal{E}_I^D & \times & \Delta Y\% & \times & Q \\ \downarrow & & \downarrow & & \downarrow \\ 0.8 & & 0.01 & & 50 = 0.4 \end{array}$$

Consider the mkt. eq.



Recall: $Q_D = a - bP$
 $Q_S = c + dP$

$$P_e = \frac{a-c}{b+d} ; Q_e = \frac{ad + bc}{b+d}$$

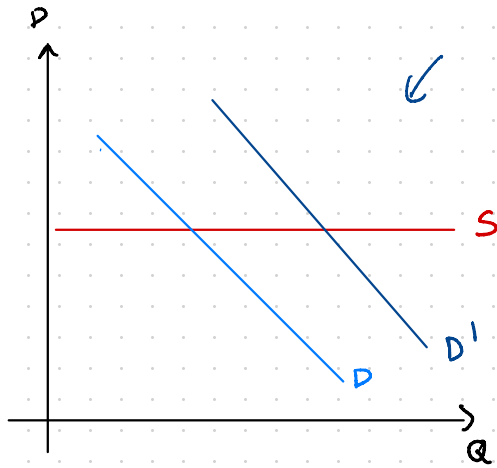
If $a \uparrow \Rightarrow$

$$Q_e = \frac{a \underline{d} + bc}{b+d} = \frac{d}{b+d} \cdot a + \frac{b}{b+d} \cdot c$$

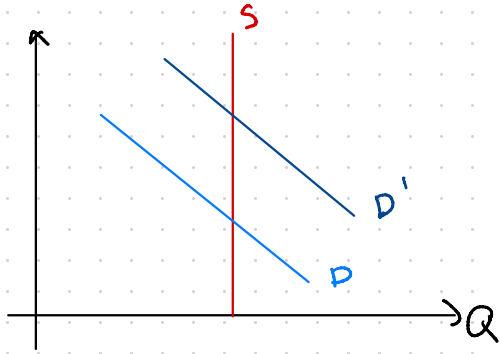
$Q_e = f(a, b, c, d)$

- $\nearrow$ slope of Q_S wrt its own price.
- $\nwarrow$ determinants of supply
- $\nearrow$ determinants of demand
- $\searrow$ slope of Q_D w.r.t. its own price

$$\frac{\Delta Q_e}{\Delta a} = \frac{d}{b+d}$$



In this case, it is enough to know
 E_{det}^D
 Q_e
 to know what will happen to



Elasticities of Supply and Demand

Point vs Arc Elasticities

$$(Q_1, P_1)$$

$$(Q_2, P_2)$$

$$\bar{P} = \frac{P_1 + P_2}{2}$$

$$\bar{Q} = \frac{Q_1 + Q_2}{2}$$

- **point elasticity of demand** Price elasticity at a particular point on the demand curve. Whenever we have accurate information about the demand schedule, we use the point elasticity of demand.

ARC ELASTICITY OF DEMAND

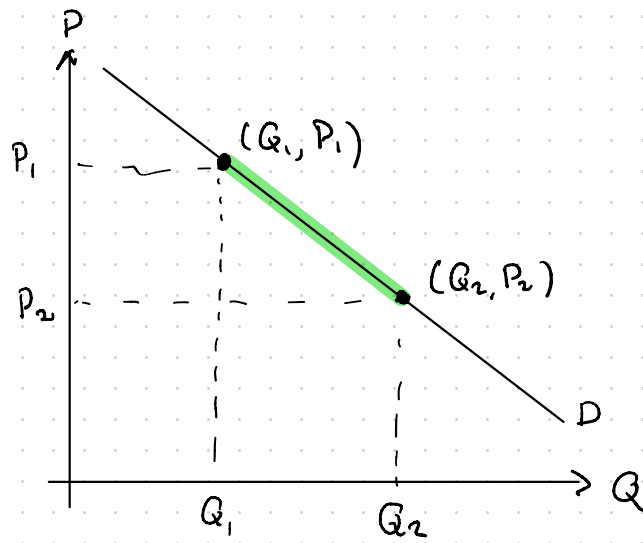
- **arc elasticity of demand** Price elasticity calculated over a range of prices. We use the arc elasticity when we only have a few points on the demand curve (two at least) but we do not have accurate information about the whole demand schedule.

$$\bar{\epsilon}_p^D = \frac{\Delta Q / \bar{Q}}{\Delta P / \bar{P}} = \frac{\Delta Q}{\Delta P} \cdot \frac{\bar{P}}{\bar{Q}}$$

- In the equation above, the bar denotes taking an average.

$$\begin{aligned} &= \frac{\frac{Q_2 - Q_1}{(Q_1 + Q_2)/2}}{\frac{P_2 - P_1}{(P_1 + P_2)/2}} = \frac{Q_2 - Q_1}{P_2 - P_1} \times \frac{\frac{P_1 + P_2}{2}}{\frac{Q_1 + Q_2}{2}} \\ &= \frac{Q_2 - Q_1}{P_2 - P_1} \times \frac{P_1 + P_2}{Q_1 + Q_2} \end{aligned}$$

Arc Elasticity:



Reasons to use the arc elasticity

- Not knowing the functional relationship of the demand (supply)
- Whenever the changes in price (determinants) are large.

A similar concept can be defined for the arc elasticity of supply as well as all the determinants.

Examples

The market for wheat

- During recent decades, changes in the wheat market had major implications for both American farmers and U.S. agricultural policy.
- To understand what happened, let's examine the behavior of supply and demand beginning in 1981.

Supply : $QS = 1800 + 240P$

Demand : $QD = 3550 - 266P$

By setting the quantity supplied equal to the quantity demanded, we can determine the market-clearing price of wheat for 1981:

$p^* = \text{eq. price}$

$$QS = QD$$

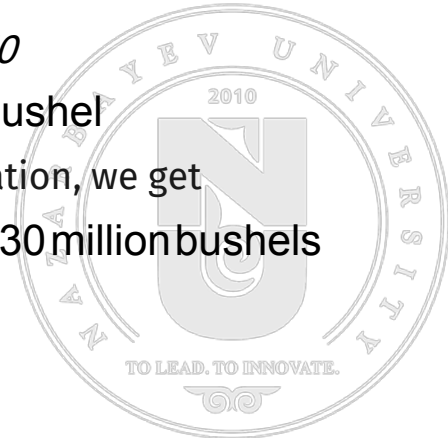
$$1800 + 240P^* = 3550 - 266P^*$$

$$506P^* = 1750$$

$$P^* = \$3.46 \text{ per bushel}$$

Substituting into the supply curve equation, we get

$$Q^* = 1800 + (240)(3.46) = 2630 \text{ million bushels}$$



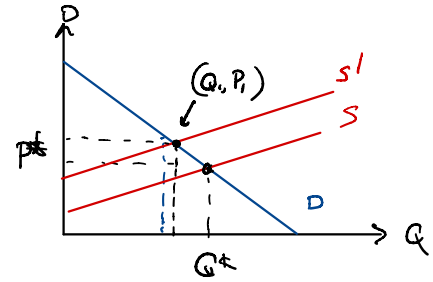
Examples

The market for wheat

$$P^* = 3.46$$
$$Q^* = 2630$$

$$\frac{\Delta Q_D}{\Delta P} = -266$$

$$\frac{\Delta Q_S}{\Delta P} = 240$$



We use the demand curve to find the price elasticity of demand: $E_P^D = \frac{P}{Q} \frac{\Delta Q_D}{\Delta P} = \frac{3.46}{2630} (-266) = -0.35$

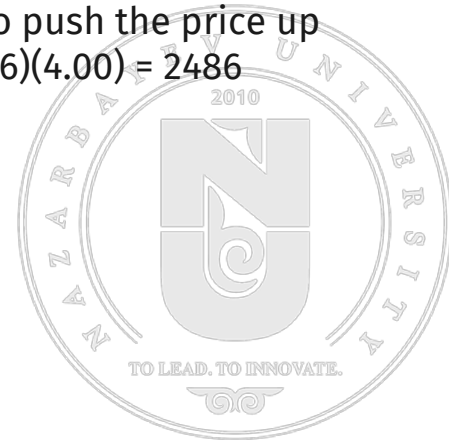
Thus demand is inelastic.

We can likewise calculate the price elasticity of supply

$$E_P^S = \frac{P}{Q} \frac{\Delta Q_S}{\Delta P} = \frac{3.46}{2630} (240) = 0.32$$

Suppose that a drought caused the supply curve to shift far enough to the left to push the price up to \$4.00 per bushel. In this case, the quantity demanded would fall to $3550 - (266)(4.00) = 2486$ million bushels. At this price and quantity, the elasticity of demand would be

$$E_P^D = \frac{4.00}{2480} (-266) = -0.43$$



Examples

The market for wheat

In 2007, demand and supply were *Demand* : $Q_D = 2900 - 125P$

$$\text{Supply} : Q_S = 1460 + 115P$$

The market-clearing (nominal) price and quantity are:

$$1460 + 115P = 2900 - 125P$$

$$P = \$3.46 \text{ per bushel}$$

$$Q = 1460 + (115)(6) = 2150 \text{ million bushels}$$

Dry weather and heavy rains, combined with increased export demand caused the price to rise considerably. You can check to see that, at the 2007 price and quantity, the price elasticity of demand was -0.35 and the price elasticity of supply 0.32 . Given these low elasticities, it is not surprising that the price of wheat rose so sharply.

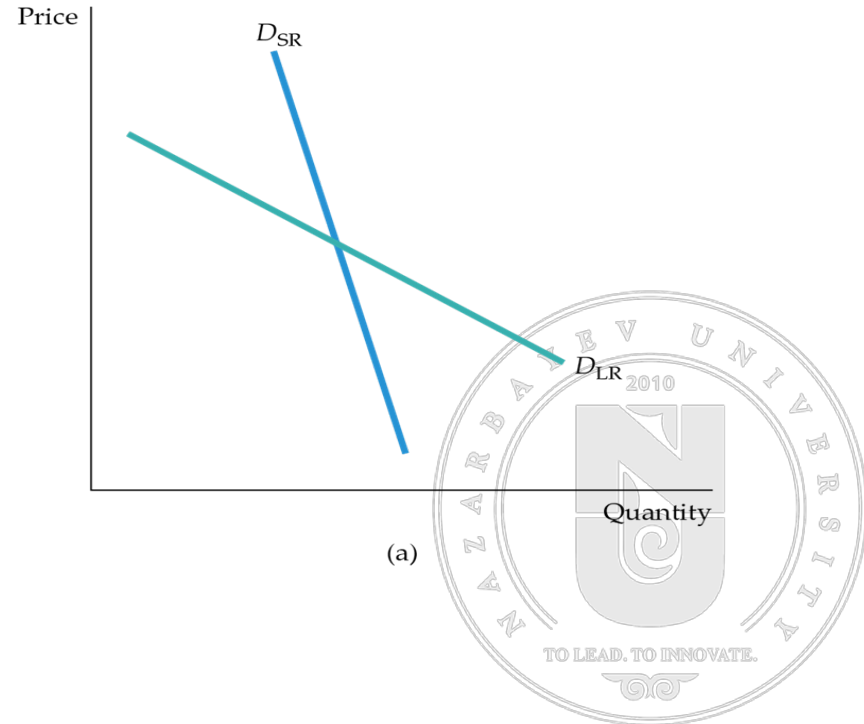


Short-run vs Long-run elasticities

(a) GASOLINE: SHORT-RUN AND LONG-RUN DEMAND CURVES

(a) In the short run, an increase in price has only a small effect on the quantity of gasoline demanded. Motorists may drive less, but they will not change the kinds of cars they are driving overnight.

In the longer run, however, because they will shift to smaller and more fuel-efficient cars, the effect of the price increase will be larger. Demand, therefore, is more elastic in the long run than in the short run.



Short-run vs Long-run elasticities

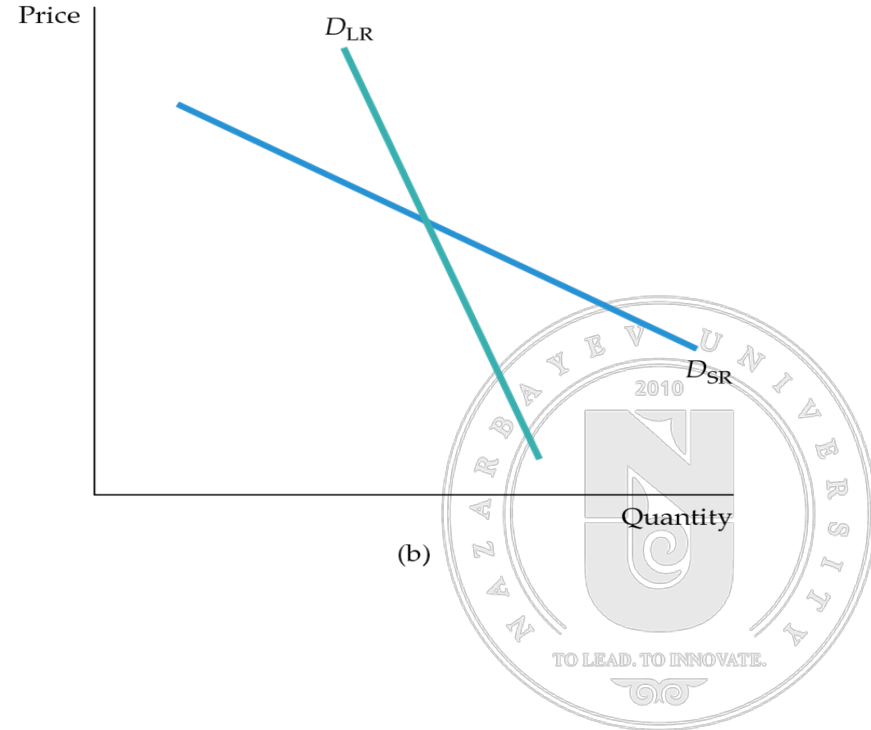
DEMAND AND DURABILITY

(b) AUTOMOBILES: SHORT-RUN AND LONG-RUN DEMAND CURVES

(b) The opposite is true for automobile demand.

If price increases, consumers initially defer buying new cars; thus annual quantity demanded falls sharply.

In the longer run, however, old cars wear out and must be replaced; thus annual quantity demanded picks up. Demand, therefore, is less elastic in the long run than in the short run.



Short-run vs Long-run elasticities

INCOME ELASTICITIES

- Income elasticities also differ from the short run to the long run.
- For most goods and services—foods, beverages, fuel, entertainment, and so on— the income elasticity of demand is larger in the long run than in the short run.
- For a durable good, the opposite is true. The short-run income elasticity of demand will be much larger than the long-run elasticity.



Short-run vs Long-run elasticities

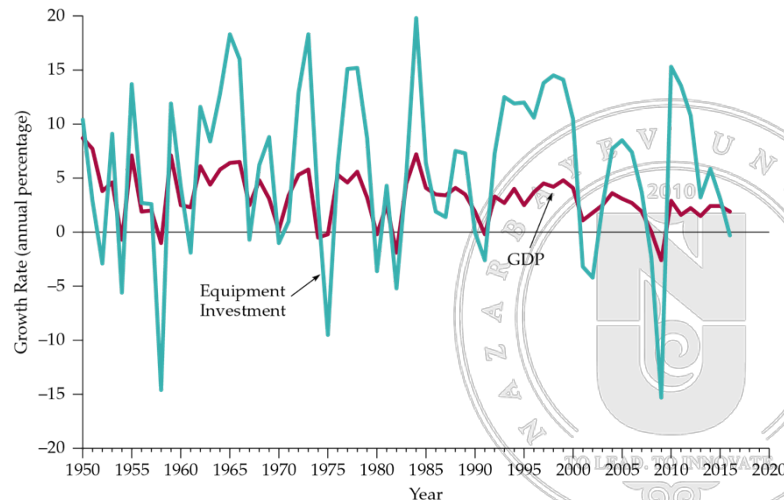
CYCLICAL INDUSTRIES

cyclical industries Industries in which sales tend to magnify cyclical changes in gross domestic product and national income.

GDP AND INVESTMENT IN DURABLE EQUIPMENT

Annual growth rates are compared for GDP and investment in durable equipment.

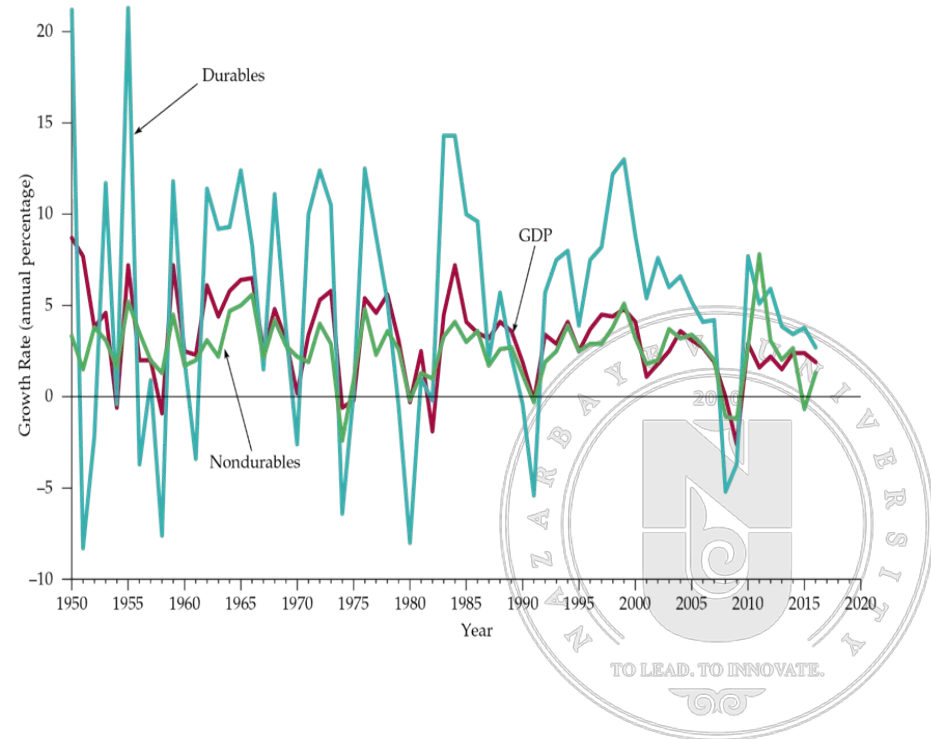
Because the short-run GDP elasticity of demand is larger than the long-run elasticity for long-lived capital equipment, changes in investment in equipment magnify changes in GDP. Thus capital goods industries are considered “cyclical.”



Short-run vs Long-run elasticities

Consumption of durables vs nondurables

- Annual growth rates are compared for GDP, consumer expenditures on durable goods (automobiles, appliances, furniture, etc.), and consumer expenditures on nondurable goods (food, clothing, services, etc.).
- Because the stock of durables is large compared with annual demand, short-run demand elasticities are larger than long-run elasticities. Like capital equipment, industries that produce consumer durables are “cyclical” (i.e., changes in GDP are magnified).
- This is not true for producers of nondurables.



Short-run vs Long-run elasticities

The demand for gasoline and automobiles

DEMAND FOR GASOLINE

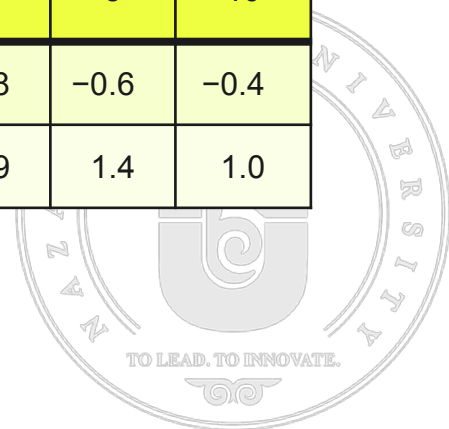
OF YEARS ALLOWED TO PASS FOLLOWING A PRICE OR

ELASTICITY	1	2	3	5	10
Price	-0.2	-0.3	-0.4	-0.5	-0.8
Income	0.2	0.4	0.5	0.6	1.0

DEMAND FOR AUTOMOBILES

OF YEARS ALLOWED TO PASS FOLLOWING A PRICE OR INCOME

ELASTICITY	1	2	3	5	10
Price	-1.2	-0.9	-0.8	-0.6	-0.4
Income	3.0	2.3	1.9	1.4	1.0



Short-run vs Long-run elasticities

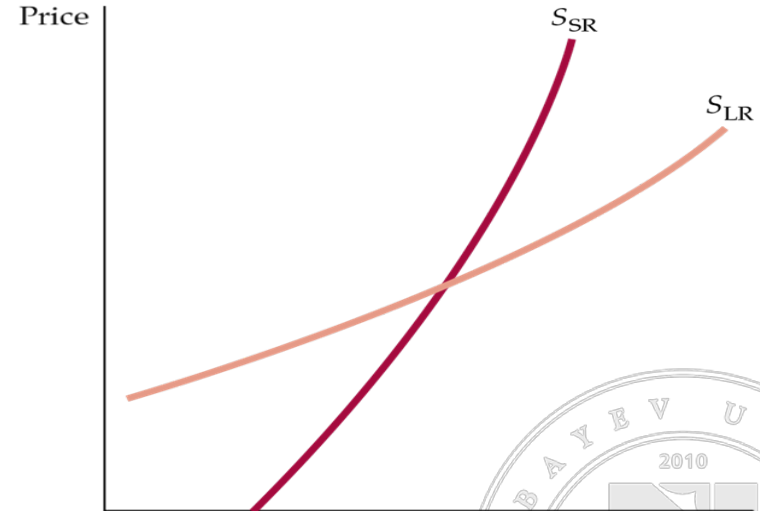
SUPPLY AND DURABILITY

Copper: short-run and long-run supply curves.

Like that of most goods, the supply of primary copper, shown in part **(a)**, is more elastic in the long run.

If price increases, firms would like to produce more but are limited by capacity constraints in the short run.

In the longer run, they can add to capacity and produce more.



(a)



Short-run vs Long-run elasticities

SUPPLY AND DURABILITY

Copper: short-run and long-run supply curves.

If the price increases, there is a greater incentive to convert scrap copper into new supply. Initially, therefore, secondary supply (i.e., supply from scrap) increases sharply.

But later, as the stock of scrap falls, secondary supply contracts.

Secondary supply is therefore less elastic in the long run than in the short run.



(b)

PRICE ELASTICITY OF:	SHORT-RUN	LONG-RUN
Primary supply	0.20	1.60
Secondary supply	0.43	0.31
Total supply	0.25	1.50

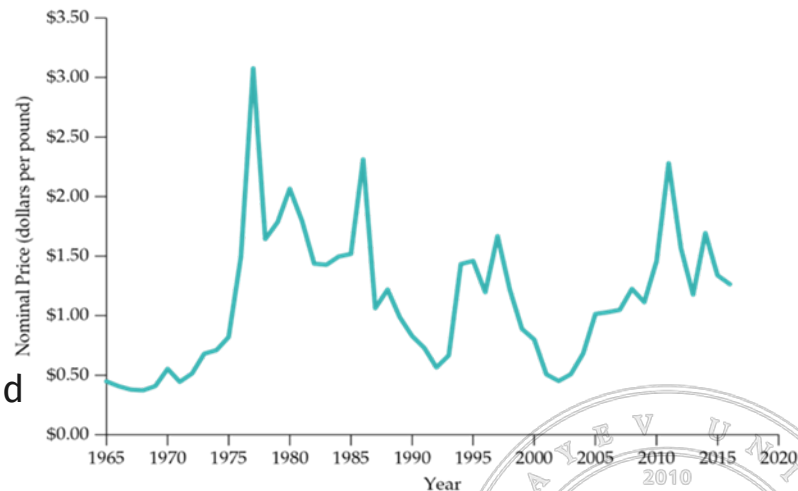
Short-run vs Long-run elasticities

The weather in Brazil and the price of coffee in New York.

PRICE OF BRAZILIAN COFFEE

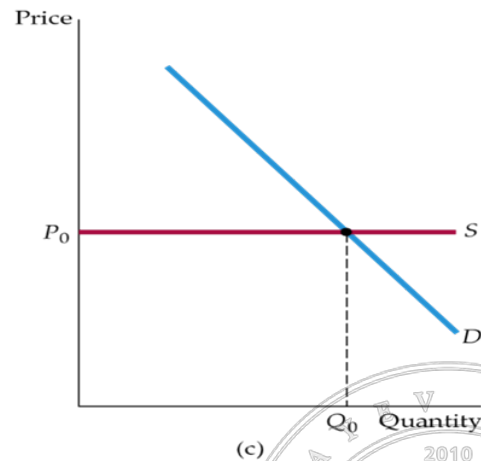
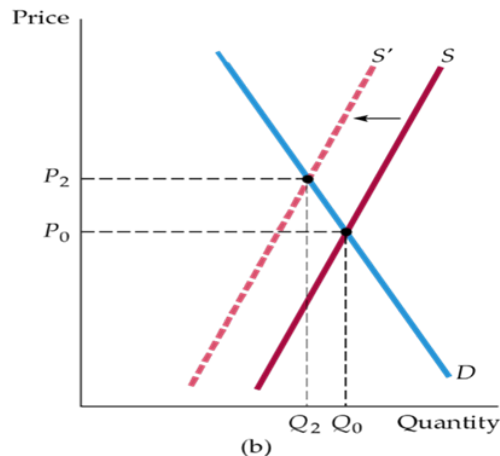
When droughts or freezes damage Brazil's coffee trees, the price of coffee can soar.

The price usually falls again after a few years, as demand and supply adjust.



Short-run vs Long-run Elasticities

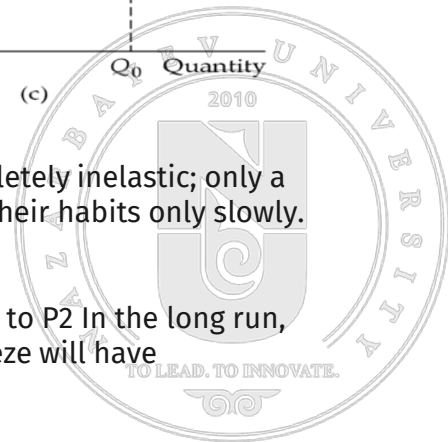
THE WEATHER IN BRAZIL AND THE PRICE OF COFFEE IN NEW YORK



SUPPLY AND DEMAND FOR COFFEE

A freeze or drought in Brazil causes the supply curve to shift to the left. In the short run, supply is completely inelastic; only a fixed number of coffee beans can be harvested. Demand is also relatively inelastic; consumers change their habits only slowly. As a result, the initial effect of the freeze is a sharp increase in price, from P_0 to P_1 .

In the intermediate run, supply and demand are both more elastic; thus price falls part of the way back, to P_2 . In the long run, supply is extremely elastic; because new coffee trees will have had time to mature, the effect of the freeze will have disappeared. Price returns to P_0 .



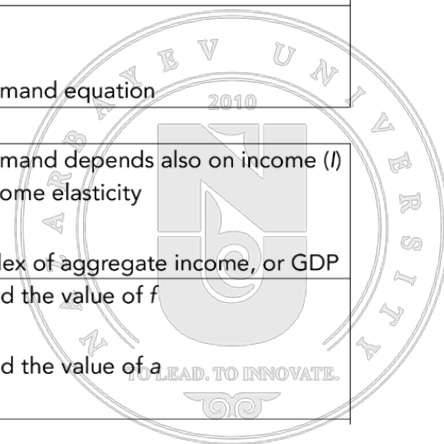
Fitting Linear supply and demand curves to data.

Our goal is to find the values of the constants a , b , c , and d in the supply and demand equations based on the known values for E_S , E_D , P^* , and Q^* . Similar procedure when demand depends also on another factor such as income.

<i>Demand: $Q = a - bP$ (2.5a)</i>	Demand equation
<i>Supply: $Q = c + dP$ (2.5b)</i>	Supply equation
<i>$E = (P/Q)(\Delta Q/\Delta P)$</i>	Elasticity
<i>For supply: $\Delta Q/\Delta P = d$</i>	Slope of supply
<i>For demand: $\Delta Q/\Delta P = -b$</i>	Slope of demand
<i>$E_D = -b(P^*/Q^*)$ (2.6a)</i>	Substitute with known values (*)
<i>$E_S = -b(P^*/Q^*)$ (2.6b)</i>	
<i>$a = Q^* + bP^*$</i>	Solve for a

<i>$Q^* = 18$ $P^* = \\$3.00$ $E_S = 1.5$ $E_D = -0.5$</i>	Known values (*)
<i>$1.5 = d(3/18) = d/6$ $d = (1.5)(6) = 9$</i>	Solve for d
<i>$18 = c + (9)(3.00) = c + 27$ $c = 18 - 27 = -9$ <i>Supply: $Q = -9 + 9p$</i></i>	Solve for c Supply equation
<i>$-0.5 = -b(3/18) = -b/6$ $18 = a = (3)(3) = a - 9$ <i>Demand: $Q = 27 - 3P$</i></i>	Demand equation

<i>$Q = a - bP + fI$ $E_I = (I/Q)(\Delta Q/\Delta I)$ $E_I = 1.3$ $I = 1.0$</i>	Demand depends also on income (I) Income elasticity Index of aggregate income, or GDP
<i>$1.3 = 1.0/18)(f)$ $f = (1.3)(18/(1.0)) = 23.4$ $b = 3; f = 23.4; P^* = 3.00; Q^* = 18$ $18 = a - (3)(3) + (23.4)(1.0) \Rightarrow a = 3.6$</i>	Find the value of f Find the value of a





End of this lecture.
Read the part on elasticity in chapter 2 of Perloff.